APPROVED 13/Jun/2025 09:27:45 GMT

APPLICATION : VP TRUCK
PROJECT : E100e / E200e

DRAWING NO
4E69000

INTERFACE

HMI TYPE : INCAB
TRACKING SYSTEM : BLUE BOX LITE
INTERFACE SYSTEM : -

VEHICLE INTERFACE

CURRENT : <150A
VOLTAGE : 12V

GRID INTERFACE

PHASES : 1PH
VOLTAGE : 230V
CURRENT : >10A
FREQUENCY : 50/60Hz

OPTION : 10/20/30/50 - ST/SP

CREATED ON : -
RESPONSIBLE ENGINEER : Jordi Garcia (JG)
DATE CHANGED : 27/03/2025
EDITOR : JG

SCHEMATICS PAGES : 5
WIRING PAGES : 1
TOTAL NO. OF PAGES : 7

A

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TITLE: COVER PAGE

DRAWN BY	MODIFIED BY	CHECKED BY	APPROVED BY	ENG APPVL BY
JGF	EC	EC	JGF	JB
DATE 10/06/2025	DATE 10/06/2025	DATE 27/03/2025	DATE 27/03/2025	DATE 27/03/2025

PROJECT:
E100e / E200e
SCHEMATIC WIRING DIAGRAM

CODE IDENT NO	SHEET SIZE	REV NO	DRAWING NO	PAGE NO
A	A3	B	4E69000	01 / 7

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AC power input
See page #3

PSU
1kW/2kW
See page #3

PSU_POS
PSU_NEG

K11
K10

BUSBAR +

OPTION: Spectrum
EF2-02 /05.4D
SS2 /05.5A
EXR2-01 /05.5C
2S-03 CFF3-1 F12 15A 2-CFF3 CF1-01 /05.1C
EF1-01 /05.1C
SS1 /05.2A
EF1-02 /05.2A
2S-01
2S-04 ACC_VEHICLE /02.6G
2S

COMP
PWR +
PWR -
eCOMPRESSOR
CHCOMP

VEHICLE

F1
100/150A
VEH_POS

E100e	100A
E200e	150A

BK-K10
BK-K11
K10_X1.5
K11_X1.5
C1-K10 -5
C1-K11 -5
2-01
2A-01
2-CFF2
2-CFF4
F11 5A
F13 5A
CFF2 -1
CFF4 -1

2-CFF1
F10 5A
CFF11
2SP

OPTION: Blue Box Lite w/ TouchLog
/02.3A 5A 12V -1
SPL 128 SHIELD-1
/02.5A GND CH /06.4G
VEHICLE SHIELD /06.5G
/02.4A 5A + Key
/02.5A -
BBL1
.1 +10..30 V DC
.7 GND
Blue Box Lite
TL
.3 KEY
.2 +
.1 -
TOUCH LOG

VEHICLE ECU
SPST / IIS active-low DO
0V when active
START /STOP SIGNAL
SPST ACC_VEHICLE /05.6E
IIS ACC_VEHICLE /05.6E
INCREASE IDLE SPEED SIGNAL

BAT+5 /02.3F
BAT+4 /02.3F

BAT+1 F25 5A BAT
IGNITION / KL15
SSW1
BAT+2 1 2 KL15 F14 5A ACC

T408
!!!

BT1

BAT8

GND /02.3F
/02.3F

VEH_NEG

BAT7 -1
/04.1E CH_BBL -7
BBL
.1 +10..30 V DC
.7 GND
Blue Box Lite

SPL113
2SPP-01 /05.6B
DCBUS-02 C1-K1-30 87-C1-K1 OROAD C20 -A8
K1 87a-C1-K1 STBY -B8
BAT1 -A7
CHDSR -A1
CH_B_01 /05.2F
C1
.A8 ACC
.B8 STD_BY1
.A7 BAT
.A1 GND
DSR3.LB1
DSR3.1 CONTROLLER

GATEWAY
ACC -E2
BAT2 -L1
DCBUS-01 -M1
2AP-01 -E4
NBAT -J3
PSUON -K2
.E2 DI_ACC
.L1 BAT
.M1 DC_BUS
.E4 DI_1
.J3 GND_BAT_SENSE
.K2 DO_6
.J1 DO_7
GATEWAY CONTROLLER
.M4 DO_3
.L4 RET_DO3
.L2 GND_CONN

2AP
SPL114
2AP-02 C1-K11 -3 X1-3 K11 X1-4 C1-K11 -4

2P
SPL117 A D1 K SPL118 2PD C1-K10 -3 X1.3 K10 X1-4 C1-K10 -4 SPL119 ISOL-02 ISOL -J1

BAT6 C1-K1 -85 K1 C1-K1 -86 ISOL-01

SPL125 D2 A K SPL126 R6 Precharging SPL127
PRECH -M4
/02.2G ACC_VEHICLE 2S-04 -L4
CHGW -L2

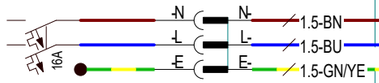
THERMO KING
Power input

SIZE	CODE IDENT NO	DWG NO	REV	SHEET
A3	-	4E69000	B	02/7

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OPTION 40/60 AC-ePTO /V2L

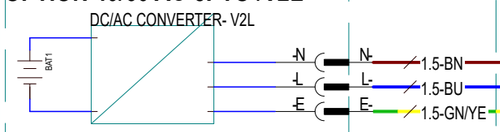
230VAC SHORE POWER

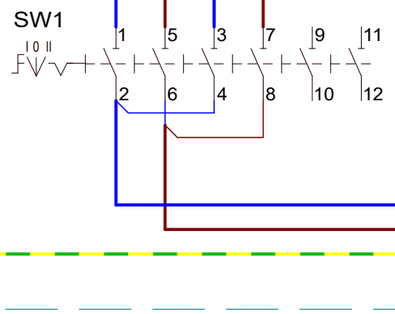


CHANGE-OVER SELECTOR

OPTION 40/60 AC-ePTO /V2L

DC/AC CONVERTER- V2L





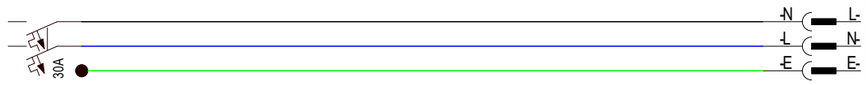
NOTES: 3 position selector SW1
1.- Positions
I: AC-ePTO / V2L
0: OFF
II: SHORE POWER
2.- 9-10 and 11-12 contacts available for dealer to signal the positions I and II respectivelly.

STANDARD OPTION 20/50

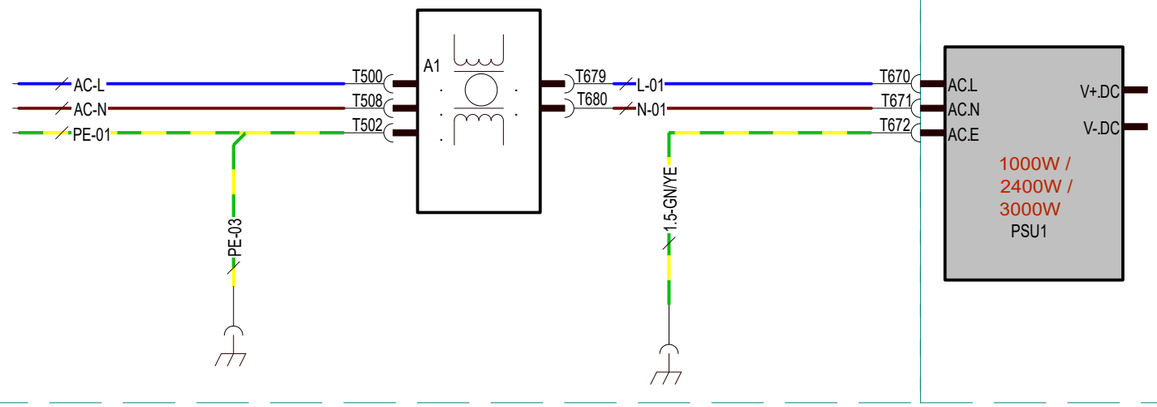
230VAC SHORE POWER

NA OPTION 20/50

230VAC SHORE POWER

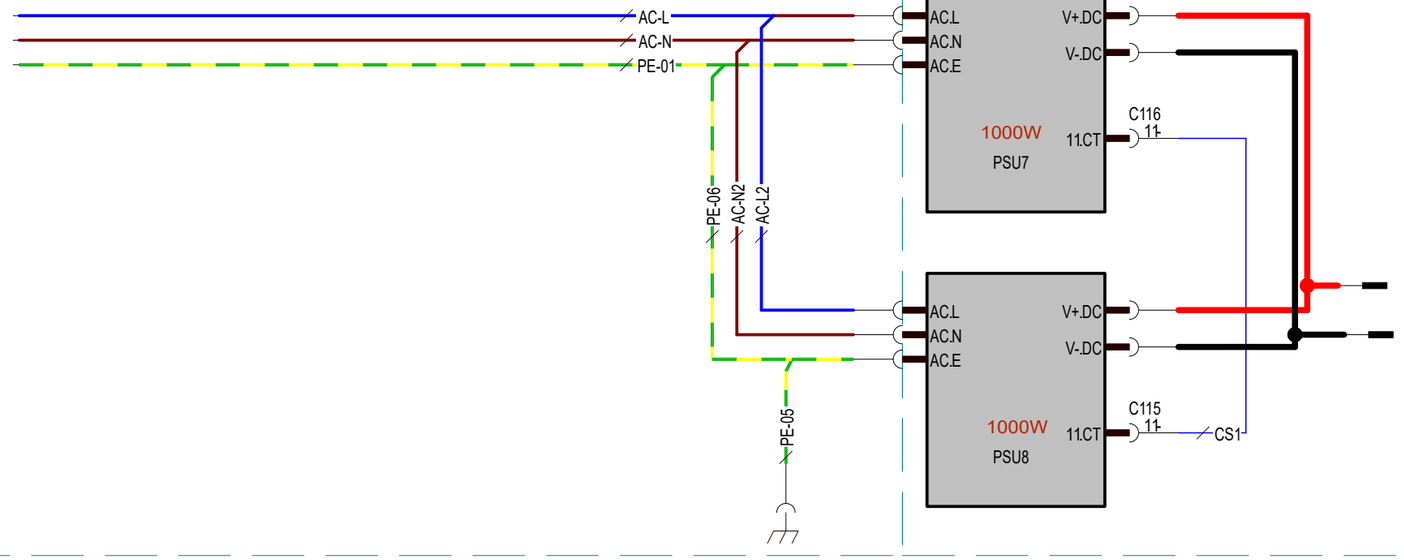


OPTION WITH FILTER



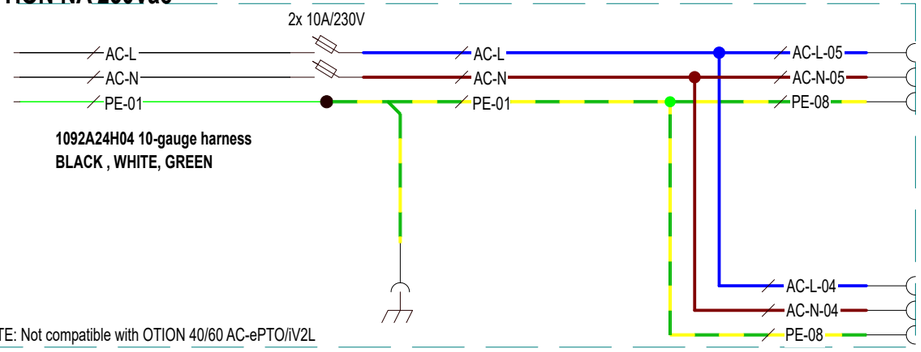
OPTION ONE POWER SUPPLY

OPTION WITHOUT FILTER



OPTION 2 POWER SUPPLIES PARALLEL OPERATION

OPTION NA 230Vac



1092A24H04 10-gauge harness
BLACK , WHITE, GREEN

NOTE: Not compatible with OTION 40/60 AC-ePTO/iV2L

THERMO KING

SIZE	CODE IDENT NO	DWG NO	REV	SHEET
A3	-	4E69000	B	03/7

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DSR3.1

DSR3.LB1 LPCO A2. — C1 C20 A2 — LPCO — C17-B — LPCO P — A-C17 — CHV; /05.1F — CH98; /02.3F — CH_BBL

DSW1 A3. — A3 — DSW1 — DSW1 1 2 — CHW; /05.2F — CH_X; /05.3F — CH_A

SGND A4. — A4 — CH5 — /04.2C; /04.2B — CH5; /04.3E — PNK; /04.4F — CH_P; /04.3E — CHT; SPL1 — CH5 — SPL3

CH5-01 in MT

SPL3 — CH_O — /04.5F

SPL3 — CHB — /04.4F

OPTION: spectrum
Only present in a Multi-Temp unit

CFi A5.

EX2 A6.

SN1 B4. — B4 — S1 — C23-A — SN1 A B — B-C23 — PNK — /04.2C

OL B2.

EX1 B3.

EFi B5.

5V C5. — C5 — 5V — C13-2 — THPCO B P A — 1-C13 — CHT — /04.2C

THPCO C4. — C4 — HP — C13-3 — THPCO C

DK1 C1. — C1 — DK1 — C16-A — ECT1 A B — B-C16 — CHB — /04.2E

DK3 C2.

SN2 C3. — C3 — S2 — C77 -A — SN2 A B — C77 B- — CH_P — /04.2C

DK4 C8.

DSW2 B1. — B1 — DSW2 — DSW2 1 2 — CH_F; /05.2F — CH_Z

OPTION: spectrum
Only present in a Multi-Temp unit

DK2 C6. — C6 — ECT2 — C78 -A — ECT2 — C78 B- — CH_O — /04.2E

GATEWAY

XG

AI_NTC G2. — G2 — CDT — C27-A — CDT A B — B-C27

RET_AI_NTC H2. — H2 — CDTN

AI_ISENSE_1 H4. — H4 — CCT — C25 -A — CCT — C25 B-

+5V_ISENSE H3. — H3 — CCT-5V

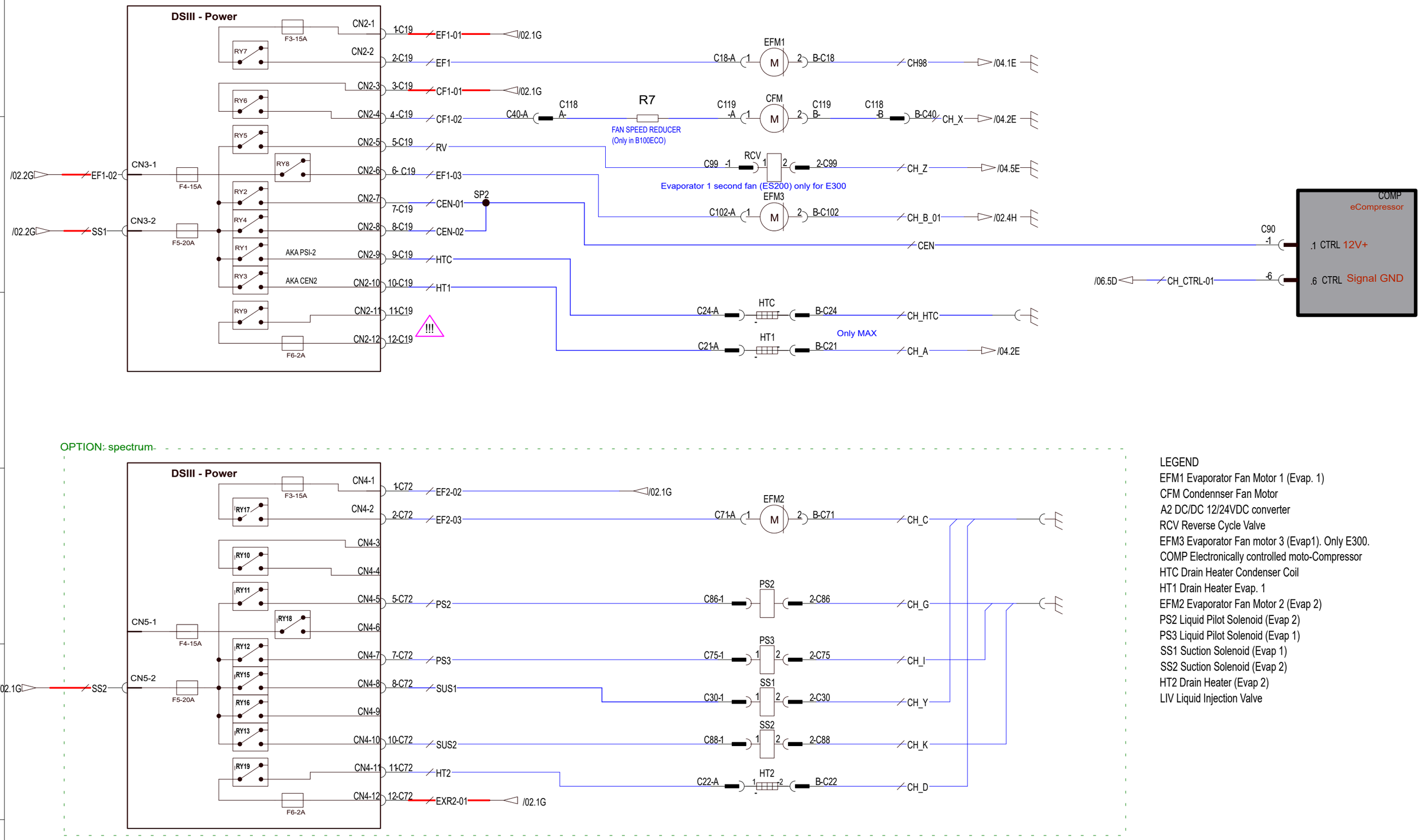
LEGEND
LPCO Low-Pressure Cut-out
DSW1 Door Switch Compartment 1
SN1 Return Air Temperature Compartment 1
THPCO High Pressure Transducer- High Pressure Cut Out
ECT1 Evaporator Coil Temperature Evap1
SN2 Return Air Temperature Compartment 2
ECT2 Evaporator Coil TEmperature Evap2
DSW2 Door Switch Compartment 2
CDT Compressor Discharge Temperature
CCT Condenser Coil Temperature

THERMO KING

DSR3.1 and Gateway Digital and analog inputs

SIZE	CODE IDENT NO	DWG NO	REV	SHEET
A3	-	4E69000	B	04/7

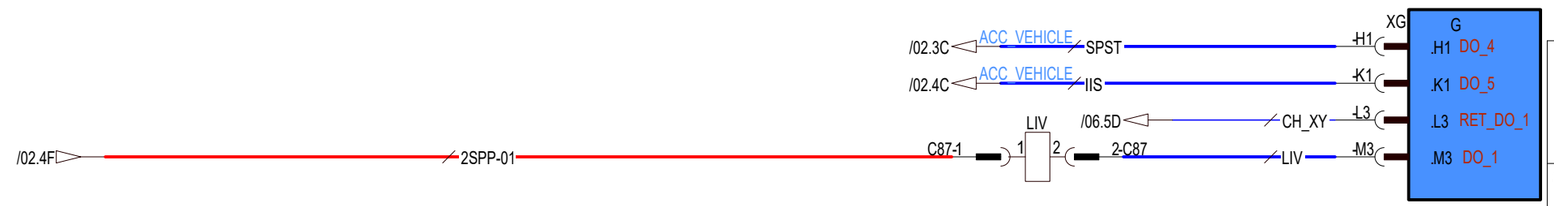
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OPTION: spectrum

LEGEND

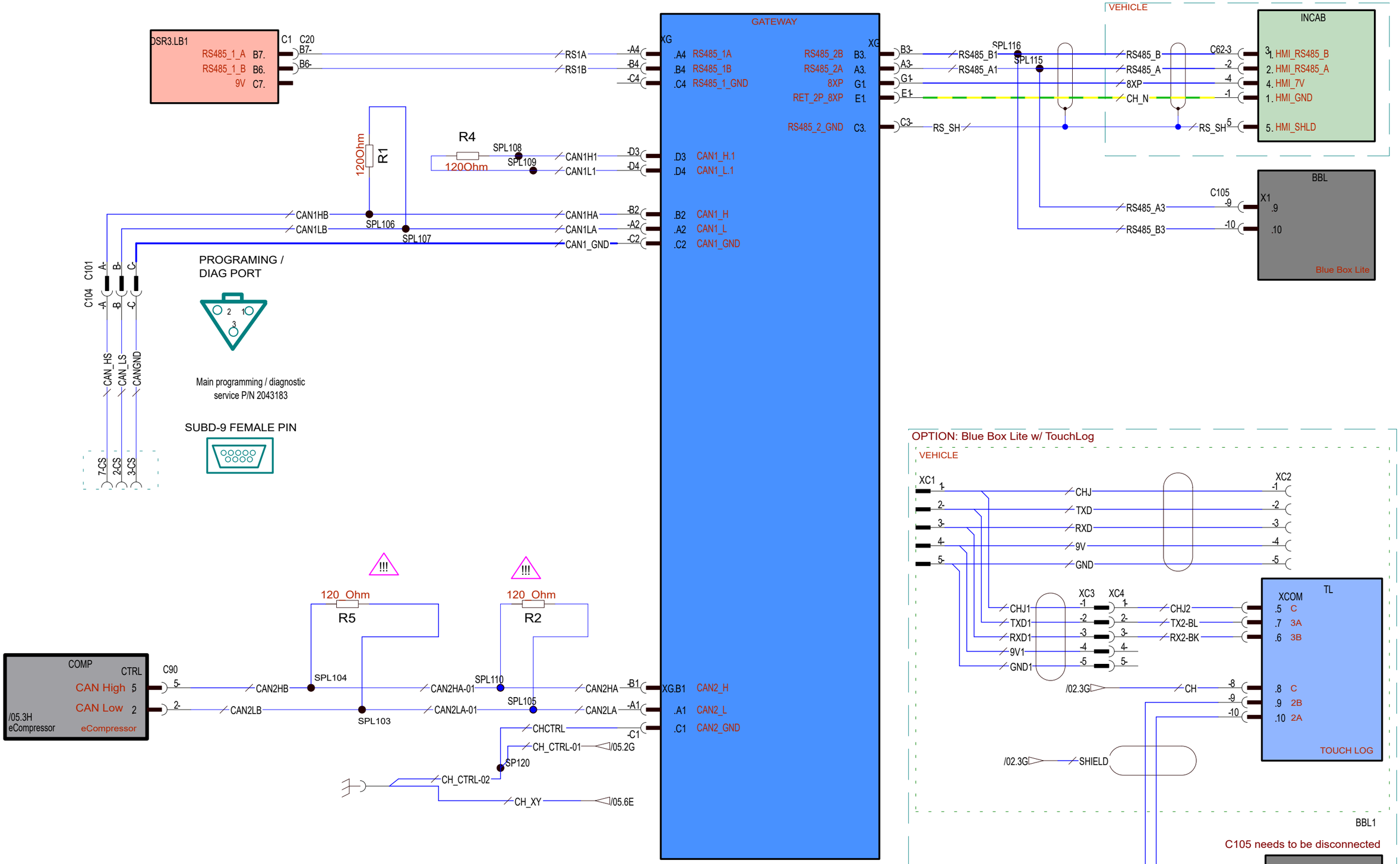
EFM1 Evaporator Fan Motor 1 (Evap. 1)
CFM Condennser Fan Motor
A2 DC/DC 12/24VDC converter
RCV Reverse Cycle Valve
EFM3 Evaporator Fan motor 3 (Evap1). Only E300.
COMP Electronically controlled moto-Compressor
HTC Drain Heater Condenser Coil
HT1 Drain Heater Evap. 1
EFM2 Evaporator Fan Motor 2 (Evap 2)
PS2 Liquid Pilot Solenoid (Evap 2)
PS3 Liquid Pilot Solenoid (Evap 1)
SS1 Suction Solenoid (Evap 1)
SS2 Suction Solenoid (Evap 2)
HT2 Drain Heater (Evap 2)
LIV Liquid Injection Valve



THERMO KING

DSRIII and Gateway Digital Outputs

SIZE	CODE IDENT NO	DWG NO	REV	SHEET
A3	-	4E69000	B	05/7



LEGEND
DSR3.1 Direct Smart Reefer 3.1 electronic Controller
In-cab HMI
GATEWAY electronic controller
BBL Blue Box Lite (Telematic device)
COMP Electronically controlled moto-Compressor
TL TouchLog. Telemetry device.

THERMO KING
Communications

SIZE	CODE IDENT NO	DWG NO	REV	SHEET
A3	-	4E69000	B	06/7

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PLUC_AC
FL1
PSU1
K11
K10
EVAPORATOR SECTION
EVAPORATOR 2 SECTION
VEHICLE
DSR3.LB1
GATEWAY
BBL1

THERMO KING

SIZE	CODE IDENT NO	DWG NO	REV	SHEET
A3	-	4E69000	B	07/7